x y

12 D At point Q where the gravitational field strength g is zero, since $g = -\frac{d\phi}{dr}$, the gravitational potential gradient must be zero.

Options A and B are wrong because the gravitational potential (and hence GPE of a mass placed there) is the largest (least negative) at Q.

Option C is wrong because the gravitational potential at a point is independent of the mass that is at that point; it is just due to the two planets.